Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: ___ZHANG Chuhan________________________________

Student ID: ___22096749D________________________________

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

__show ip route ospf_________________ 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/0__________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___129______ 2

Does R2 show up as an OSPF neighbor on R1? ___Yes_____ 2

Does R2 show up as an OSPF neighbor on R3? ___No_____ 2

What does this information tell you?

All traffic from R3 to the 192.168.2.0/24 network will be routed via R1. The S0/0/1 interface on R2 remains configured as a passive interface, so OSPF routing information is not advertised across this interface. The total cost of 129 arises because traffic from R3 to the 192.168.2.0/24 network must traverse two T1 (1.544 Mb/s) serial links (each with a cost of 64) along with the R2 Gigabit 0/0 LAN link (with a cost of 1). 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1____________________ 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ___S0/0/1_________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 (one T1 (1.544 Mb/s) serial link with a cost of 64 plus the R2 Gigabit 0/0 LAN link with a cost of 1).

Is R2 listed as an OSPF neighbor to R3? __Yes______

2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Modern networking equipment supports link speeds exceeding 100 Mb/s. To achieve more accurate cost calculations for these higher-speed links, a higher default reference-bandwidth value is required.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24:   R1 S0/0/1 + R3 G0/0 (781+1=782).

The cost to 192.168.23.0/30: R1 S0/0/1 + R3 S0/0/1 (781+64=845).

2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1,562.

Each serial link now has a cost of 781, and the route to the 192.168.23.0/24 network passes through two serial links. So 781 + 781 = 1,562.

2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF selects the route with the lowest accumulated cost. The path with the smallest accumulated cost is R1-S0/0/0 + R2-S0/0/1 + R3-G0/0, which equals 781 + 781 + 1 = 1,563. This value is lower than the accumulated cost of R1-S0/0/1 + R3-G0/0, which is 1565 + 1 = 1,566.

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   Router ID assignments determine the election of the Designated Router (DR) and Backup Designated Router (BDR) on a multi-access network. If the router ID is tied to an active interface, it may change if that interface goes down. Therefore, it should be configured using the IP address of a loopback interface (which does not go down) or explicitly set with the router-id command.

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2. Why is the DR/BDR election process not a concern in this lab?

   The DR/BDR election process only applies to multi-access networks such as Ethernet or Frame Relay. The serial links used in this lab are point-to-point links, so no DR/BDR election occurs.

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3. Why would you want to set an OSPF interface to passive?

   Configuring a LAN interface as passive removes unnecessary OSPF routing information from that interface, thereby conserving bandwidth. The router will still advertise the network to its neighbors.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of OSPF areas makes the protocol hierarchical and allows very scalable network design.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP route will be used, because route selection between different routing protocols is based on administrative distance, not the protocol metric. EIGRP (internal) has AD 90, which is lower than OSPF 110 and RIP 120.

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